

Ê PHASOR TEMPLATES (Air: ε_r=μ_r=1)

Wavenumber per medium <i>nonmag.</i>			
$k_i = \frac{\omega\sqrt{\epsilon_{r,i}}}{c}$ (rad/m) $\Rightarrow k_2 = k_1\sqrt{\epsilon_{r,2}/\epsilon_{r,1}}$			
Direction table $u \in \{x, y, z\}$ = axis perp. to boundary			
Inc dir	Inc	Refl	Trans
+û	e^{-jk_1u}	e^{+jk_1u}	e^{-jk_2u}
-û	e^{+jk_1u}	e^{-jk_1u}	e^{+jk_2u}

+û direction ⇔ Med 2 at u ≥ 0. Reflected sign flips, transmitted matches incident.

General template (incident has pol. ê, amplitude a ₀):		
$\tilde{\mathbf{E}}^i = a_0 \hat{e} e^{\mp j k_1 u}$ (V/m)	(incident wave, Med 1)	
$\tilde{\mathbf{E}}^r = \Gamma a_0 \hat{e} e^{\pm j k_1 u}$ (V/m)	(reflected wave, Med 1)	
$\tilde{\mathbf{E}}^t = \tau a_0 \hat{e} e^{\mp j k_2 u}$ (V/m)	(transmitted wave, Med 2)	
(lossy med 2: replace $e^{\mp j k_2 u}$ with $e^{\mp \alpha_2 u} e^{\mp j \beta_2 u}$)		
Total in medium 1: $\tilde{\mathbf{E}}_1 = \tilde{\mathbf{E}}^i + \tilde{\mathbf{E}}^r$.		
ê by polarization plug into templates above		
Pol.	ê	Note
Lin. x̂	ŷ	E _y = 0
Lin. ŷ	ŷ	E _x = 0
Lin. at 45°	(x̂ + ŷ)/√2	in phase
RHC	x̂ - jŷ	δ = -π/2
LHC	x̂ + jŷ	δ = +π/2
General	a _x x̂ + a _y e ^{jδ} ŷ	elliptical

Γ, τ act on ê unchanged. Only swap the polarization, not the formulas.

TABLE 8-2 — Γ, τ, R, T (unitless) SUMMARY

<i>Med. 1 (z < 0) incident side; Med. 2 (z ≥ 0) transmitted.</i>			
	⊥ pol (θ _i = 0)	pol	
Γ refl. wav	$\frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$	$\frac{\eta_2 \cos \theta_i - \eta_1 \cos \theta_t}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t}$	$\frac{\eta_2 \cos \theta_t - \eta_1 \cos \theta_i}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i}$
τ trans. wav	$\frac{2\eta_2}{\eta_2 + \eta_1}$	$\frac{2\eta_2 \cos \theta_i}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t}$	$\frac{2\eta_2 \cos \theta_t}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i}$
τ rel.	τ = 1 + Γ	τ _⊥ = 1 + Γ _⊥	τ = (1 + Γ) $\frac{\cos \theta_i}{\cos \theta_t}$
R refl. pwr	Γ ²	Γ _⊥ ²	Γ ²
T trans. pwr	τ ² $\frac{\eta_1}{\eta_2}$	τ _⊥ ² $\frac{\eta_1 \cos \theta_t}{\eta_2 \cos \theta_i}$	τ ² $\frac{2 \eta_1 \cos \theta_t}{\eta_2 \cos \theta_i}$
R + T total	T = 1 - R	T _⊥ = 1 - R _⊥	T = 1 - R
Notes: sin θ _t = √μ ₁ ε ₁ /μ ₂ ε ₂ sin θ _i ; nonmag. ⇒ η ₂ /η ₁ = n ₁ /n ₂ .			
Intrinsic impedance η _i plug into table above			
η _i = √(μ _r /ε _r) η ₀ $\xrightarrow{\mu_r=1}$ $\frac{\eta_0}{\sqrt{\epsilon_{r,i}}}$ (Ω) η ₀ = 120π ≈ 377 Ω			
Default μ _r = 1 (air, dielectric). Use μ _r ≠ 1 only if problem says "ferromagnetic"/iron/ferrite/etc.			
Low-loss correction η _c σ/(ωε) ≪ 1			
η _c ≈ √(μ _r /ε _r) (1 + j $\frac{\sigma}{2\omega\epsilon'}$) $\xrightarrow{\text{drop } j}$ $\frac{\eta_0}{\sqrt{\epsilon_r}}$			
For Γ/τ just use η ₀ /√ε _r . The j term is for η _c , θ _η . See LOSSY MEDIA — 5 CASES.			

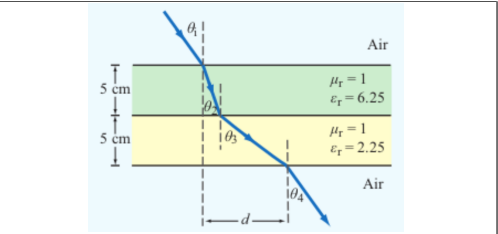
POWER REFLECTION & TRANSMISSION

% reflected <i>always</i> %P _r = 100 Γ ²
% transmitted <i>η-ratio matters!</i> %P _t = 100 τ ² η ₁ /η ₂
Check: %P _r + %P _t = 100.

SNELL'S LAW — OBLIQUE INCIDENCE

Single boundary <i>angles from normal</i>	$\sin \theta_2 = \sin \theta_1 \sqrt{\frac{\epsilon_{r1}}{\epsilon_{r2}}} = \sin \theta_1 \frac{n_1}{n_2}$
Multilayer chain <i>stack of slabs 1 → 2 → 3 → 4</i>	$\sin \theta_{i+1} = \sin \theta_i \sqrt{\frac{\epsilon_{r,i}}{\epsilon_{r,i+1}}}$
Air-slabs-air shortcut <i>outer media identical</i>	θ _{exit} = θ _{incident}
<i>Snell only sets angles. Reflection/transmission magnitudes need Fresnel coefficients (parallel vs perpendicular).</i>	

LATERAL DISPLACEMENT



Beam offset through N slabs thickness t_i, refracted angle inside slab i Slab-indexed θ_i = angle inside slab i; t_i = slab thickness (m)

$$d = \sum_{i=1}^N t_i \tan \theta_i$$

The angle (θ_i) in the formula is always the one the ray makes inside that slab of height (t_i).

LOSSY MEDIA — 5 CASES

Loss tangent ε' = ε _r ε ₀ ; ε ₀ = 8.854 × 10 ⁻¹² F/m		
	ε''/ε'	σ/ωε ₀
Type	σ/ωε	η _c , α, β
Lossless (σ = 0)	0	η = η ₀ /√ε _r exact α = 0, β = ω√ε _r /c
Low-loss	< 10 ⁻²	η _c ≈ $\frac{\eta_0}{\sqrt{\epsilon_r}}$ (1 + j $\frac{\sigma}{2\omega\epsilon'}$) α ≈ $\frac{\sigma\eta_0}{2\sqrt{\epsilon_r}}$, β ≈ $\frac{\omega\sqrt{\epsilon_r}}{c}$
Quasi-cond.	10 ⁻² –10 ²	exact: (η/√X)e ^{jθ_η} see below
Good cond.	> 10 ²	η _c ≈ (1 + j)√πfμ/σ α ≈ β ≈ √πfμσ
Magnetic	any	keep full √μ _r /ε _r η ₀
For Γ/τ: drop j correction; use η ₀ /√ε _r . Magnetic: keep μ _r .		
Low-loss quick params σ/(ωε) ≪ 1, μ _r = 1		
α ≈ $\frac{\sigma\eta_0}{2\sqrt{\epsilon_r}}$ (Np/m), β ≈ $\frac{\omega\sqrt{\epsilon_r}}{c}$ (rad/m), η _c ≈ $\frac{\eta_0}{\sqrt{\epsilon_r}}$ (Ω)		
Exact (quasi-cond) X = √(1 + (ε''/ε') ²)		
α = ω√(μ _r ε'/2)√X-1, β = ω√(μ _r ε'/2)√X+1		
θ _η = 1/2 arctan σ/ωε'		

For Γ/τ: drop j correction; use η₀/√ε_r. Magnetic: keep μ_r.

Low-loss quick params σ/(ωε) ≪ 1, μ_r = 1
α ≈ $\frac{\sigma\eta_0}{2\sqrt{\epsilon_r}}$ (Np/m), β ≈ $\frac{\omega\sqrt{\epsilon_r}}{c}$ (rad/m), η_c ≈ $\frac{\eta_0}{\sqrt{\epsilon_r}}$ (Ω)
Exact (quasi-cond) X = √(1 + (ε''/ε')²)
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θ_η = 1/2 arctan σ/ωε'

RECTANGULAR WAVEGUIDE — MODES

TE mode E_z = 0, H_z ≠ 0 TM mode H_z = 0, E_z ≠ 0
Dimensions a × b with a > b along x̂, ŷ; wave in ẑ.
E_x form (TE) read m, n from arguments
E_x ∝ cos($\frac{m\pi x}{a}$) sin($\frac{n\pi y}{b}$) sin(ωt - βz)
coef. on x = mπ/a; coef. on y = nπ/b.

Identify TE_{mn} vs TM_{mn} cos/sin pattern of E_x is identical for both!

1. Read the problem statement (e.g. “a TE wave” → TE).
2. If m = 0 or n = 0 in the mode label ⇒ must be TE (TM forbids 0).
3. Source field H_z(x, y) given ⇒ TE. Source field E_z(x, y) given ⇒ TM.

TE allows at least one of m, n ≥ 1 (other can be 0). TM needs both m, n ≥ 1. So TE₁₀ exists but TM₁₀ doesn't.

TABLE 8-3 — FULL FIELD COMPONENTS

e^{-jβz} factor omitted. k_c² = (mπ/a)² + (nπ/b)². Ê in V/m, Ĥ in A/m, Z in Ω.

TE mode <i>generator: Ê_z</i>	
$\tilde{H}_z = H_0 \cos \frac{m\pi x}{a} \cos \frac{n\pi y}{b}$	
$\tilde{E}_x = \frac{j\omega\mu}{k_c^2} \left(\frac{n\pi}{b}\right) H_0 \cos \frac{m\pi x}{a} \sin \frac{n\pi y}{b}$	
$\tilde{E}_y = -\frac{j\omega\mu}{k_c^2} \left(\frac{m\pi}{a}\right) H_0 \sin \frac{m\pi x}{a} \cos \frac{n\pi y}{b}$	
$\tilde{E}_z = 0, \tilde{H}_x = -\tilde{E}_y/Z_{TE}, \tilde{H}_y = \tilde{E}_x/Z_{TE}$	
$Z_{TE} = \eta/\sqrt{1 - (f_c/f)^2}$	
TM mode <i>generator: Ê_z</i>	
$\tilde{E}_z = E_0 \sin \frac{m\pi x}{a} \sin \frac{n\pi y}{b}$	
$\tilde{E}_x = -\frac{j\beta}{k_c^2} \left(\frac{m\pi}{a}\right) E_0 \cos \frac{m\pi x}{a} \sin \frac{n\pi y}{b}$	
$\tilde{E}_y = -\frac{j\beta}{k_c^2} \left(\frac{n\pi}{b}\right) E_0 \sin \frac{m\pi x}{a} \cos \frac{n\pi y}{b}$	
$\tilde{H}_z = 0, \tilde{H}_x = -\tilde{E}_y/Z_{TM}, \tilde{H}_y = \tilde{E}_x/Z_{TM}$	
$Z_{TM} = \eta\sqrt{1 - (f_c/f)^2}$	
TEM plane wave <i>for reference</i>	
$\tilde{E}_x = E_0 e^{-j\beta z}, \tilde{H}_y = \tilde{E}_x/\eta$	
$\eta = \sqrt{\mu/\epsilon}(\Omega), f_c \text{ N/A}, k = \omega\sqrt{\mu\epsilon}$	
Properties common to TE/TM	
$f_c = \frac{u_{p0}}{2} \sqrt{(m/a)^2 + (n/b)^2}$ (Hz)	
$\beta = k\sqrt{1 - (f_c/f)^2}$ (rad/m)	
$u_p = \frac{\omega}{\beta} = \frac{u_{p0}}{\sqrt{1 - (f_c/f)^2}}$ (m/s)	

CUTOFF FREQUENCY

Common cutoffs (a > b) f _c formula in Table 8-3	
Mode	f _{mn}
u _{p0}	c/√ε _r (hollow: u _{p0} = c)
TE ₁₀	f ₁₀ = u _{p0} /(2a) (dominant)
TE ₀₁	f ₀₁ = u _{p0} /(2b)
TE ₂₀	f ₂₀ = u _{p0} /a = 2f ₁₀
TE ₁₁ , TM ₁₁	f ₁₁ = $\frac{u_{p0}}{2} \sqrt{1/a^2 + 1/b^2}$
Propagation requires f > f _{mn} . In Z _{TE} /Z _{TM} and u _g , f _c = f _{mn} of the excited mode.	

ε_r FROM DISPERSION

Given ω, β, mode (m, n) result is unitless	
$\epsilon_r = \frac{c^2}{\omega^2} \left[\beta^2 + \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 \right]$	(unitless)
Inputs: ω (rad/s), β (rad/m), a, b (m), c = 3 × 10 ⁸ (m/s), m, n (unitless).	

GROUP VELOCITY & TRAVEL TIME

u_g = u_{p0}√(1 - (f_{mn}/f)²) (m/s), u_pu_g = u_{p0}²
t = L/u_g (s) travel time through guide of length L
Higher modes have lower u_g ⇒ arrive later.

BOUNDARY

η_i, θ_t — imp. (Ω)
Γ, τ, R, T — coeffs (unitless)
τ = 1 + Γ; R = |Γ|²
k_i = ω√ε_{r,i}/c (rad/m)
α₂ (Np/m), β₂ (rad/m): lossy
z = 0 — boundary plane (m)
η₀ = 120π ≈ 377 Ω

SNELL / OBLIQUE

θ_i, θ_t — angles from normal
n_i = √ε_{r,i} — index of refr.
t_i — slab thickness (m)
d — lateral disp. (m)
plane of inc. = plane of k+normal
⊥: **E** ⊥ this plane
||: **E** || this plane

WAVEGUIDE

a, b — wide, narrow dim. (m);
a > b
m, n — mode indices (int.)
f_{mn} — cutoff (Hz)
β — prop. const (rad/m)
Z_{TE}/TM — wave imp. (Ω)
E_x — E-field comp. (V/m)
H_y — H-field comp. (A/m)
η = η₀/√ε_r

VELOCITIES

u_{p0} = c/√ε_r — bulk phase vel.
u_p = u_{p0}/√(1 - (f_c/f)²)
u_g = u_{p0}√(1 - (f_c/f)²)
u_pu_g = u_{p0}²
t = L/u_g — travel time
c = 3 × 10⁸ m/s

POWER & UNITS

%P_r, %P_t — pwr refl./trans. (unitless)
%P_r = 100|Γ|²
%P_t = 100|τ|² η₁/η₂
%P_r + %P_t = 100
σ (S/m), ε (F/m), μ (H/m)
ε₀ = 8.85 × 10⁻¹² F/m
μ₀ = 4π × 10⁻⁷ H/m

DIPOLE TYPE BY LENGTH

STEP 0 — always start here $\lambda = \text{wavelength (m)}, l = \text{antenna rod length (m)}, c = 3 \times 10^8 \text{ m/s}$

$\lambda = \frac{c}{f} \text{ (m)}$	ratio $\frac{l}{\lambda} \rightarrow$	pick row below
l/λ	Type	Use formula
$< 1/50$	Hertzian (short)	HERTZIAN $S(R, \theta)$, $D = 1.5$
$= 1/4$	Quarter-wave	ARB formula, $\pi l/\lambda = \pi/4$
$= 1/2$	Half-wave	half-wave shortcut, $D = 1.64$, $R_{\text{rad}} = 73 \Omega$
$= 1$	Full-wave	ARB, $D = 2.41$, $R_{\text{rad}} = 199 \Omega$
other	Arbitrary	ARB formula

If $l/\lambda < 1/50$ STOP and use Hertzian. Do NOT plug small l into the arbitrary formula — wrong shape.

HERTZIAN DIPOLE — $S(R, \theta)$

Far-field E,H phasors small $l \ll \lambda$
$\vec{E}_\theta = \frac{jI_0 k l \eta_0}{4\pi R} e^{-jkR} \sin \theta$, $\vec{H}_\phi = \vec{E}_\theta / \eta_0$
Power density $l/\lambda < 1/50$; far field
$S(R, \theta) = \frac{\eta_0 k^2 I_0^2}{32\pi^2 R^2} \sin^2 \theta = S_{\text{max}} \sin^2 \theta \text{ (W/m}^2\text{)}$
l rod length (m); I_0 peak current (A); R obs. dist. (m); θ from dipole axis; $k = 2\pi/\lambda$ (rad/m); $\eta_0 = 120\pi \approx 377 \Omega$.
Max @ broadside $\theta_{\text{max}} = \pi/2$; D exact $D = 1.5$ (1.76 dB)
Total radiated power $P_{\text{rad}} = \frac{\eta_0 \pi}{3} (I_0 l/\lambda)^2 \text{ (W)}$

ARBITRARY-LENGTH DIPOLE — $S(\theta)$

Power density at R any l
$S(\theta) = \frac{15 I_0^2}{\pi R^2} \left[\frac{\cos(\frac{\pi l}{\lambda} \cos \theta) - \cos(\frac{\pi l}{\lambda})}{\sin \theta} \right]^2$
At $\theta = 0, \pi$: bracket is 0/0. L'Hôpital $\Rightarrow S = 0$ (no axial radiation). TI-36X Pro alt: evaluate bracket at $\theta = 0.001$ rad $\rightarrow$ tiny $\rightarrow 0$.
Normalized pattern dimensionless $F(\theta) = S(\theta)/S_{\text{max}}$
Quarter-wave $l = \lambda/4$ $\pi l/\lambda = \pi/4$, $\cos(\pi/4) = \sqrt{2}/2$
$\frac{S(\theta)}{S_0} = \left[\frac{\cos(\frac{\pi}{4} \cos \theta) - \frac{\sqrt{2}}{2}}{\sin \theta} \right]^2$, $S_0 = \frac{15 I_0^2}{\pi R^2}$
$S_{\text{max}} = S(\pi/2) = S_0 \left(1 - \frac{\sqrt{2}}{2}\right)^2 \approx 0.0858 S_0$
$D = 1.64$, $R_{\text{rad}} = 36.5 \Omega$, $\theta_{\text{max}} = \pi/2$.
Half-wave $l = \lambda/2$ $\pi l/\lambda = \pi/2$, shortcut form
$S(\theta) = \frac{15 I_0^2}{\pi R^2} \frac{\cos^2(\frac{\pi}{2} \cos \theta)}{\sin^2 \theta}$
$D = 1.64$, $R_{\text{rad}} = 73 \Omega$, $\theta_{\text{max}} = \pi/2$.

COMMON ANGLES (deg $\leftrightarrow$ rad, trig values)

Conversion multiply by $\pi/180$ or $180/\pi$
$\theta_{\text{rad}} = \theta_{\text{deg}} \cdot \frac{\pi}{180}$ $\theta_{\text{deg}} = \theta_{\text{rad}} \cdot \frac{180}{\pi}$
$^\circ$ rad $\sin \theta$ $\cos \theta$ $\sin^2 \theta$ $\cos^2 \theta$
0 0 0 1 0 1
30 $\pi/6$ 1/2 $\sqrt{3}/2$ 1/4 3/4
45 $\pi/4$ $\sqrt{2}/2$ $\sqrt{2}/2$ 1/2 1/2
60 $\pi/3$ $\sqrt{3}/2$ 1/2 3/4 1/4
90 $\pi/2$ 1 0 1 0
180 π 0 -1 0 1

Antenna formulas usually take θ in rad. RAD mode on calc.

DIPOLE

l — antenna length (m)
 $\lambda = c/f$ (m); $c = 3 \times 10^8 \text{ m/s}$
 I_0 — peak current (A)
 $k = 2\pi/\lambda$ (rad/m)
 R — obs. dist. (m); θ from axis
 $\eta_0 = 120\pi \approx 377 \Omega$
 $S(R, \theta)$ — pwr density (W/m²)

BEAM

$F(\theta) = S/S_{\text{max}}$ (unitless)
 β — HPBW (rad or $^\circ$)
 Ω_p — pattern solid angle (sr)
 $D = 4\pi/\Omega_p$ — directivity
 ξ — rad. efficiency (0–1)
 $G = \xi D$ — gain (unitless)
 $D_{\text{dB}} = 10 \log_{10} D$

FRIIS / dB

P_t, P_r — TX/RX power (W)
 G_t, G_r — linear gains
 $S_r = G_t P_t / (4\pi R^2)$ (W/m²)
 $P_r = G_t G_r (\lambda/4\pi R)^2 P_t$
 $G = 10^{G_{\text{dB}}/10}$
 $3 \text{ dB} = \times 2$, $10 \text{ dB} = \times 10$
 $P_N = k_B T_{\text{sys}} B$ (W)

APERTURE

A_p — physical area (m²)
 $A_e = \lambda^2 D / (4\pi)$ (m²)
 $A_e \approx A_p$ for aperture
 $D \approx 4\pi A_p / \lambda^2$ (unitless)
 $\beta \approx 0.88 \lambda / \ell$ (rect/sq.)
 $\beta \approx 1.02 \lambda / d$ (circular)
 $2A_p \Rightarrow 2D$, $\beta/\sqrt{2}$

ARRAY

N — number of elements
 d — spacing (m)
 $\gamma = kd \cos \theta$
 $F_a = \sin^2(N\gamma/2) / \sin^2(\gamma/2)$
 $F_a^{\text{max}} = N^2$ at $\theta = \pi/2$
 $F_a^{\text{norm}} = F_a / N^2$
 $\beta \approx 0.88 \lambda / (Nd)$ (broadside)

BEAM PARAMETERS — β, Ω_p, D

Step 1 — Normalize always start here
$F(\theta) = S(\theta)/S_{\text{max}}$ (unitless)
Step 2 — HPBW β half-power beamwidth, full angular width at -3 dB
Set $F(\theta) = 0.5$, solve for $\theta \rightarrow \theta_{1/2}$
$\beta = 2\theta_{1/2}$ (symmetric beams, rad or $^\circ$)
Gaussian:
$F(\theta_{1/2}) = e^{-a\theta_{1/2}^2} = 0.5 \Rightarrow \theta_{1/2} = \sqrt{\ln 2/a}$
Step 3 — Pattern solid angle Ω_p units: sr
$\Omega_p = \int_0^{2\pi} \int_0^\pi F(\theta) \sin \theta d\theta d\phi$
No ϕ dep.: $\Omega_p = 2\pi \int_0^\pi F(\theta) \sin \theta d\theta$.
TI-36X Pro: [2nd] [d/dx] for $\int_0^\pi$, set RAD mode, multiply result by 2π .
Narrow-beam Gaussian shortcut ($\sin \theta \approx \theta$, $\rightarrow \infty$):
$\Omega_p \approx \frac{\pi}{a} = \frac{\pi \theta_{1/2}^2}{\ln 2}$ ($F = e^{-a\theta^2}$)

Step 4 — Directivity D
$D = \frac{4\pi}{\Omega_p}$ (unitless) $D_{\text{dB}} = 10 \log_{10} D$
Alt. with two orthogonal HPBW's: $D \approx 4\pi / (\beta_{xz} \beta_{yz})$.
Step 5 — Gain (if asked) $\xi = \text{rad. eff.}$, $0 < \xi \leq 1$
$G = \xi D$ $G_{\text{dB}} = 10 \log_{10} G$
Lossless / ideal antenna $\Rightarrow G = D$.

COMMON DIPOLE PARAMETERS

Type	l	D	D_{dB}	R_{rad} (Ω)
Isotropic	N/A	1	0	—
Hertzian	$< \lambda/50$	1.5	1.76	$80\pi^2 (l/\lambda)^2$
Half-wave	$\lambda/2$	1.64	2.15	73.2
$\lambda/4$	$\lambda/4$ (gnd)	1.64	2.15	36.6
monopole				
Full-wave	λ	2.41	3.82	199

Half-wave: $P_{\text{rad}} = 36.6 I_0^2 W$. $\lambda/4$ monopole (above ground): same patt. as half-wave but $P_{\text{rad}}, R_{\text{rad}}$ halved.
 Lossless antenna ($\xi = 1$) $\Rightarrow G = D$. So in Friis, use $G_t, G_r =$ these D values (e.g. half-wave $G = 1.64$).

EFFECTIVE AREA A_e

Definition antenna's RX cross section (m²)
$A_e = \frac{\lambda^2 D}{4\pi} \text{ (m}^2\text{)}$
Physical cross section wire dipole, diameter d
$A_p = l d = \frac{\lambda}{2} d$ (half-wave)
Inverse: D from A_e
$D = \frac{4\pi A_e}{\lambda^2}$ (unitless)
Hertzian special case $D = 1.5$
$A_e^{\text{Hertz}} = \frac{3\lambda^2}{8\pi} \approx 0.119 \lambda^2 \text{ (m}^2\text{)}$
Thin wire: $A_e \gg A_p$. Antenna captures field over $\sim \lambda$.

LINEAR ANTENNA ARRAY

Setup N identical elements along axis, spacing d
elem pos $z_i = id$, $i = 0, \dots, N-1$; $A_i = a_i e^{j\psi_i}$
Equal phase ($\psi_i = 0$) $\Rightarrow$ broadside beam ($\theta_{\text{max}} = \pi/2$).
Progressive phase $\psi_i = i\alpha \Rightarrow$ steered beam at $\cos \theta_{\text{max}} = -\alpha/(kd)$.
Pattern multiplication S_e = single element pattern
$S(R, \theta, \phi) = S_e(R, \theta, \phi) F_a(\theta)$
Array factor (general) $k = 2\pi/\lambda$
$F_a(\theta) = \left \sum_{i=0}^{N-1} A_i e^{jkid \cos \theta} \right ^2$
Uniform $A_i = 1$, equal phase closed form
$F_a(\theta) = \frac{\sin^2(N\gamma/2)}{\sin^2(\gamma/2)}$, $\gamma = kd \cos \theta$
Peak: $F_a^{\text{max}} = N^2$ at $\gamma = 0$ (broadside, $\theta = \pi/2$). Normal-ize: $F_a^{\text{norm}} = F_a/N^2$.
HPBW (broadside, uniform) use Nd, not $(N-1)d$
$\beta \approx 0.88 \frac{\lambda}{Nd}$ (rad) $= 50.8^\circ \cdot \frac{\lambda}{Nd}$
Grating lobes appear if $d > \lambda$ (broadside). Avoid.

FRIIS TRANSMISSION FORMULA

Pwr density at RX, then RX pwr G_t, G_r linear gains; $\lambda = c/f$

$$S_r = \frac{G_t P_t}{4\pi R^2} \text{ (W/m}^2\text{)}; \quad P_r = G_t G_r \left(\frac{\lambda}{4\pi R} \right)^2 P_t \text{ (W)}$$

Equivalent: $P_r = S_r A_{e,r}$.

Solve for R max range

$$R = \frac{\lambda}{4\pi} \sqrt{\frac{G_t G_r P_t}{P_r}} \text{ (m)}$$

Equivalent form using A_e recu. area form

$$P_r = \frac{G_t P_t A_{e,r}}{4\pi R^2}, \quad G_r = \frac{4\pi A_{e,r}}{\lambda^2}$$

Free-space path loss: $L_{fs} = (4\pi R/\lambda)^2$, so $P_r = G_t G_r P_t / L_{fs}$.

General form (off-axis) when patterns NOT peak-aligned

$$P_r = G_t G_r \left(\frac{\lambda}{4\pi R} \right)^2 P_t F_t(\theta_t, \phi_t) F_r(\theta_r, \phi_r)$$

F_t, F_r normalized patterns at the link directions; $F = 1$ when aligned.

Recipe 1. $\lambda = c/f$. 2. Convert dB gains to linear: $G = 10^{G_{\text{dB}}/10}$.

3. If TX is “half-wave dipole” use $G_t = 1.64$. 4. Plug into Friis.

Use linear G , not dB. Convert FIRST.

dB $\leftrightarrow$ LINEAR

Works for any power-like quantity: G gain, D directivity, P_r/P_t , SNR, etc.

$$X_{\text{dB}} = 10 \log_{10} X_{\text{lin}} \quad X_{\text{lin}} = 10^{X_{\text{dB}}/10}$$

$$\text{e.g. } D_{\text{dB}} = 10 \log_{10} D; \quad G_{\text{dB}} = 10 \log_{10} G; \quad \text{SNR}_{\text{dB}} = 10 \log_{10} (P_r/P_N)$$

dB linear	dB linear
0	1
3	2
6	4
7	5
−3	0.5
−10	0.1

Memorize: $3 \text{ dB} \Leftrightarrow \times 2$, $10 \text{ dB} \Leftrightarrow \times 10$. Add dB = multiply linear.

For E-field / amplitude use $20 \log$ not $10 \log$ (power $\propto E^2$).

NOISE & SNR

Noise power thermal noise into matched receiver
$P_N = k_B T_{\text{sys}} B$ (W)
$k_B = 1.38 \times 10^{-23} \text{ J/K}$; T_{sys} system noise temp (K); B receiver bandwidth (Hz).
Signal-to-noise ratio
$\text{SNR} = \frac{P_{\text{rec}}}{P_N}$ (unitless) $\text{SNR}_{\text{dB}} = 10 \log_{10} \frac{P_{\text{rec}}}{P_N}$
Typical comm. link needs $\text{SNR} > 10 \text{ dB}$ for usable signal.

APERTURE ANTENNAS (horn, dish)

Directivity (any shape, uniform illum.)

$$D \approx \frac{4\pi A_p}{\lambda^2} \text{ (unitless)} \Rightarrow A_e \approx A_p$$

Directivity from beamwidths *any aperture*

$$D \approx \frac{4\pi}{\beta_{xz}\beta_{yz}} \text{ (use rad); circular: } D \approx 4\pi/\beta^2$$

HPBW and D by aperture shape *uniform illum.; β in rad*

Shape	A_p	HPBW (rad)	D (unitless)
Rect. $\ell_x \times \ell_y$	$\ell_x \ell_y$	$\beta_x \approx 0.88\lambda/\ell_x$ $\beta_y \approx 0.88\lambda/\ell_y$	$D \approx 4\pi/(\beta_x \beta_y)$ $= 4\pi \ell_x \ell_y / \lambda^2$
Square ℓ	ℓ^2	$\beta \approx 0.88\lambda/\ell$	$D \approx 4\pi/\beta^2 = 4\pi \ell^2 / \lambda^2$
Circular (diam. d)	$\pi d^2/4$	$\beta \approx 1.02\lambda/d$	$D \approx 4\pi/\beta^2 = \pi^2 d^2 / \lambda^2$

Scaling rules $D \propto A_p/\lambda^2$; $\beta \propto \lambda/\sqrt{A_p}$		
Change	New D	New β
Double ($A_p \rightarrow 2A_p$)	area $D_{\text{new}} = 2D_{\text{old}}$	$\beta_{\text{new}} = \beta_{\text{old}}/\sqrt{2}$
Double freq ($\lambda \rightarrow \lambda/2$)	$D_{\text{new}} = 4D_{\text{old}}$	$\beta_{\text{new}} = \beta_{\text{old}}/2$